Digital Communication IC Design

Homework #2

(Due on 2026/10/21)

- Please upload your codes and reports (with simulation results and required explanations) to the NTUCool.
- Please disclose how you use AI tools and its version.

Total points: 130 points

1. Use the prime sequence length $N = 53$. Let $u = 1 + (m \bmod 52)$, where m is the last two digits of the student ID. For odd N, define the root- u Zadoff-Chu sequence as

$$x_u[n] = e^{-j\pi un(n+1)/N}, n = 0, \dots, N-1$$

(a) Generate $x_u[n]$. Plot its real part, imaginary part, magnitude, and unwrapped phase. (10%)

(b) Calculate and stem-plot the normalized periodic autocorrelation for $l = 0,1,\dots N-1$. (10%) Also calculate a nonperiodic linear autocorrelation and explain why it does not exhibit the same zero-sidelobe property. (10%)

$$R_u[l] = \frac{1}{N}\sum_{n=0}^{N-1} x_u[n]{x_u}^*[(n-l) \bmod N]$$

$$R_u^{\text{lin}}[l] = \frac{1}{N}\sum_{n=l}^{N-1} x_u[n]x_u^*[n-l].$$

(c) Choose $v = 1 + ((u + 16) \bmod 52)$. Calculate the periodic cross-correlation $C_{u,v}[l]$ for $l = 0,1,\dots N-1$. (10%)

$$C_{u,v}[l] = \frac{1}{N}\sum_{n=0}^{N-1} x_u[n]{x_v}^*[(n-l) \bmod N]$$

(d) Construct a student-specific synchronization test with $u_0 = 3 + ((7m + 2) \bmod 48)$, $\tau_0 = (5m + 3) \bmod 48$, complex gain $\alpha = exp(j\pi/5)$. The received signal is defined as

$$r[n] = \alpha x_{u_0}[(n-\tau_0) \bmod N], \qquad n = 0,1,\dots,N-1,$$

If the set $\Omega = \{u_0 - 2, u_0, u_0 + 2\}$, search three possible roots in set Ω and 53 cyclic delays using the metric below. For each root in the set Ω, draw $D(u,\tau)$ versus τ (15%). Report the detected root and delay, and whether they match u_o and τ_o. **Please add some**

explanations (5%)

$$D(u,\tau) = \left|\frac{1}{N}\sum_{n=0}^{N-1} r[n]x_u^*[(n-\tau)\ mod\ N]\right| \text{ for } u \in \Omega \text{ and } \tau = 0,1,\ldots 52$$

2. Given the Barker code list in Table 1, please check the periodic autocorrelation function defined as

$$\Phi_s(k) = \frac{1}{N}\sum_{n=0}^{N-1} S(n)S^*((n-k) \bmod N)$$

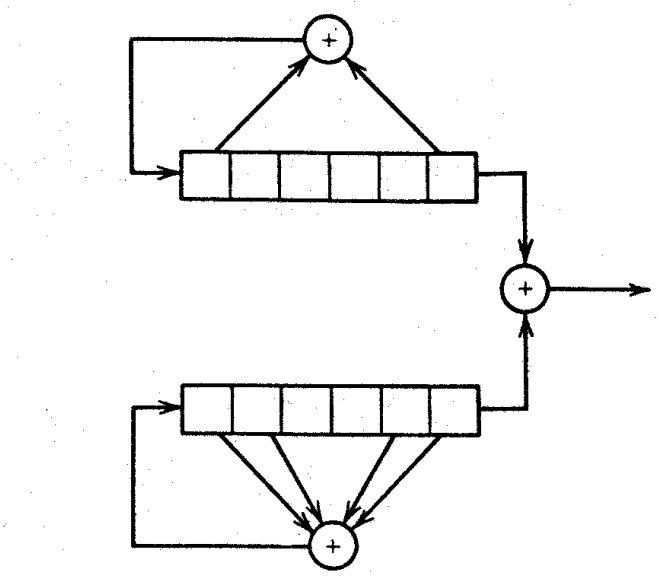
for $N = 7$ and 13. (15%)

Table 1 Barker code

N	Barker code
7	$[-1\ \ -1\ \ -1\ \ 1\ \ 1\ \ -1\ \ 1]$
11	$[-1\ \ -1\ \ -1\ \ 1\ \ 1\ \ 1\ \ -1\ \ 1\ \ 1\ \ -1\ \ 1]$
13	$[-1\ \ -1\ \ -1\ \ -1\ \ -1\ \ 1\ \ 1\ \ -1\ \ -1\ \ 1\ \ -1\ \ 1\ \ -1]$

3. The octal primitive polynomials 103_8 and 147_8 form a degree-6 preferred pair and correspond to $p(x) = x^6 + x + 1$ and $p'(x) = x^6 + x^5 + x^2 + x + 1$. Let $r_1 = 1 + (m\ mod\ 20)$ and $r_2 = 11 + (m\ mod\ 20)$. Convert r_1 and r_2 to six-bit nonzero initial states.

(a) Connect your LFSR as in the following figure.



Set the initial states r_1 into the top LFSR and set all 0's as the initial state in the bottom

LFSR. Generate the Maximal-length sequences $s_1[n]$ of 63 bits, for $n = 0,1,\ldots,62$. Print the initial values, the complete sequences. Then, Set the initial states r_2 into the bottom LFSR and set all 0's as the initial state in the top LFSR. Generate the Maximal-length sequences $s_2[n]$ of 63 bits, for $n = 0,1,\ldots,62$. Print the initial values, the complete sequences. (10%)

(b) Map a binary sequence $s_i[n]$ to $d_i[n] = (-1)^{s_i[n]}$. For both sequences, verify the normalized periodic autocorrelation for all 63 lags. (20%)

$$R_i[l] = \frac{1}{63}\sum_{n=0}^{62} d_i[n]d_i[(n+l)\ mod\ 63]$$

(c) Now, we will construct another codeword of the Gold code. Put initial state r_1 and r_2 into top LFSR and bottom LFSR, respectively. Generate the output $s_3[n]$ of 63 bits. Now, map $s_3[n]$ to $d_3[n]$. Then, let's check the cross-correlation property for $(u,v) = (1,3)$ and $(u,v) = (2,3)$ with $N = 63$. (20%)

$$C_{u,v}[l] = \frac{1}{N}\sum_{n=0}^{N-1} d_u[n]d_v[(n-l)\ mod\ N]$$

(d) Please judge if your answer in (c) is correct or not? Please state the reason. (5%)